

Print Estimating & Order Management System – Project Brief

Project Overview

We are developing a **custom, hybrid print estimating and order management system** for internal company use. The software must support **macOS** as a **standalone application** with **manual cloud sync**, and also be available as a **web-based cloud application** for remote access.

The system should be clean, fast, flexible, and highly **customizable** — capable of supporting **multiple printing disciplines** like **wide-format signage, digital commercial printing, folded packaging, and flexographic printing**.

Objectives

- **Standalone macOS Desktop App** for in-house use.
 - **Cloud-Based Web App** for remote access.
 - **Manual or scheduled sync** between local and cloud.
 - **User-configurable workflows** tailored to different print types.
 - High-performance estimating engine with dynamic pricing.
 - Fast, clean, intuitive interface.
-

User Roles

1. **Admin**
 - Full access to all system functions, configurations, and users.
 2. **Sales Representative**
 - Access to customers, estimates, jobs, order tracking.
 3. **Production Staff**
 - Limited access to jobs in production, packing slips, and CoCs.
-

Key Features

Login & Company Selection

- Login screen with secure authentication.
- Upon login, user selects the **company account** (e.g., “Premier Printing”, “SignFX”).

Dashboard

- Central navigation hub to access:
 - Customers
 - Prospects
 - Estimating
 - Sales Reps
 - Jobs
 - Inventory
 - Vendors

Customer & Prospect Management

- Customer database with full order/estimate history.
- View, duplicate, and edit previous orders.
- Assign sales rep to customer.
- Convert prospect to customer.

Industry-Specific Workflows

When setting up a company or job, the user selects an **industry type**:

- **Flexographic Printing**
 - Die database
 - Stepping/imposition
 - Ink usage
- **Digital Commercial Printing**
 - Simple estimating
 - Shrink wrap, binding, folding
- **Wide Format Signage**
 - Grommets, pole pockets, seaming

 *All industry workflows are fully customizable.*

Estimating System

- Select print type and inputs based on workflow.
- User-configurable estimating components:
 - Materials
 - Inks
 - Machinery

- Labor
 - Finishing options
 - Accurate cost calculations and markup logic.
 - Branded PDF generation of estimates.
 - Convert estimate to job.
-

Order Management

- Status tracking: Estimate → Approved → Production → Complete → Shipped
 - Generate:
 - Job tickets
 - Packing slips
 - Certificates of Conformance (CoCs)
-

Customization Tools

Admins can:

- Add/Edit print equipment and assign costs
 - Define estimating fields per industry
 - Add materials, labor, finishing options
 - Adjust pricing logic and defaults per job type
 - Configure workflows without needing a developer
-

Sync Mechanism

- Manual or scheduled data sync from local SQLite (desktop app) to cloud PostgreSQL (web app).
 - Conflict handling rules: timestamps + version tracking.
 - Sync includes:
 - Estimates
 - Jobs
 - Inventory
 - Customers
 - Configs
-

Deliverables

- UI Design (Login, Dashboard, Forms, Estimate Builder, etc.)

- Working macOS app with local database
 - Cloud-based admin panel
 - API for sync
 - Complete documentation (user manual + developer guide)
 - Source code and deployment instructions
-

References & Inspirations

- [PrintPLANR](#) – UI & estimating approach
- [LabelTraxx](#) – Flexo workflow
- [PrintLogic](#) – Digital commercial estimating
- [Billmade](#) – Order/document flow